

## Review

# Assessing the Feasibility of Repurposing the Existing Natural Gas Pipelines for Hydrogen Transport—A Comprehensive Review

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## Abstract

In a bid to investigate the optimum transportation method for offshore wind-produced hydrogen ( $H_2$ ) and assess the feasibility of repurposing the existing oil and gas infrastructure for  $H_2$  transmission, this paper assesses the existing  $H_2$  transportation methods with a comprehensive review of the  $H_2$  impact on the existing natural gas pipeline infrastructure. To establish the possibility of repurposing the existing natural gas (NG) pipelines for  $H_2$  gas transport, this paper reviews the influential technical measures—composition, pressure, temperature, volumetric energy density, density, and pressure drop—to assess whether the characteristics of hydrogen gas are compatible with the natural gas pipeline infrastructure. Based on these reviews, it was found that the current NG pipeline pressure exacerbates the  $H_2$  embrittlement; for the existing NG pipelines to be repurposed, the operating pressure should be reduced, and the pipeline material should be revised. It was found that higher strength steels can be re-used with major modifications, or the pipeline should be constructed from material grade X52 or below. Nevertheless, the fitness of the existing NG pipelines for  $H_2$  transmission should be assessed on a case-by-case basis and other factors such as erosion, leakage, pressure cycling, monitoring (e.g., distributed fiber-optic sensing technology) and a rigorous assessment of welds and joints should also be considered.

**Keywords:** hydrogen ( $H_2$ );  $H_2$  gas transmission; gas pipelines; natural gas transmission; pipelines repurpose; pipelines operating parameters; pipeline material; transmission costs

## 1. Introduction

Hydrogen is viewed as a practical solution to completely decarbonize the global economy and is therefore gaining momentum. Hydrogen is very important because it leaves zero-carbon footprint when deployed for energy consumption [1]. The last decade saw the emergence of fossil fuel as the dominant source of our energy needs [2]. Currently, increasing population, urbanization, and economy are placing increased demands on energy needs, driving the implementation of renewable energy sources like wind [3]. Climate is another factor driving the switch to cleaner alternatives [4]. To support the increased integration of renewables, like offshore wind, the utilization of energy storage solutions like the offshore production of green  $H_2$  from the excess in wind energy becomes essential [5]. Electrolyzers are used to generate  $H_2$  from excess in offshore wind energy [6] and this  $H_2$  can then be transported using pipelines. Therefore, an opportunity exists in deploying the existing oil and gas offshore infrastructure for the transportation of this offshore generated  $H_2$ . It has been found that repurposing an existing pipeline to transport hydrogen instead of constructing an entirely new pipeline is considerably cheaper [7]. Jens



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et al. (2021) [8] estimated that the cost of repurposing an existing pipeline to transmit hydrogen is approximately €0.2 to €0.6 M/km, compared to around €1.4 to €3.4 M/km to build a new pipeline [7,8].

Currently, few hydrogen pipelines exist around the world. In Belgium, there is an 80 km long pipeline with an internal diameter (ID) of 5.9 inches which operates at 100 barg [9]. In England, there is a 16 km long pipeline which operates at 50 barg. Another example is a 220 km long hydrogen pipeline in Germany that operates at 20 barg and has linked Dusseldorf and Recklinghausen for 60 years. It has an internal diameter of 3.9 inches to 11.8 inches [9] and transports 1,000,000 m<sup>3</sup> of H<sub>2</sub> per year. Other examples include a 550 km long pipeline in France with an internal diameter (ID) of 100 mm transporting 2,000,000 m<sup>3</sup> of H<sub>2</sub> yearly, a 100 km H<sub>2</sub> pipeline laid in Texas, and hydrogen pipelines covering several kilometers in Iowa, Louisiana and Alberta, Canada [9].

Considering that oil and gas production is on the decline and in due course will be replaced by offshore renewables such as wind, the transmission of the hydrogen gas generated from offshore renewable systems (like wind) using the stranded oil and gas infrastructure presents a promising opportunity.

This paper investigates the optimal methods of offshore hydrogen transportation and assesses the viability of reusing the existing natural gas pipeline infrastructure for H<sub>2</sub> transport. A thorough literature review was conducted and all H<sub>2</sub> transmission options were evaluated based on their pros and cons. Based on this, H<sub>2</sub> transport in pipelines was identified as the optimal solution for the transportation of H<sub>2</sub> over short to medium distances.

The feasibility of reusing the existing NG pipelines for the transport of hydrogen gas was also reviewed in this paper using technical measures like composition, pressure, temperature, pressure drop, volumetric energy density and operating density.

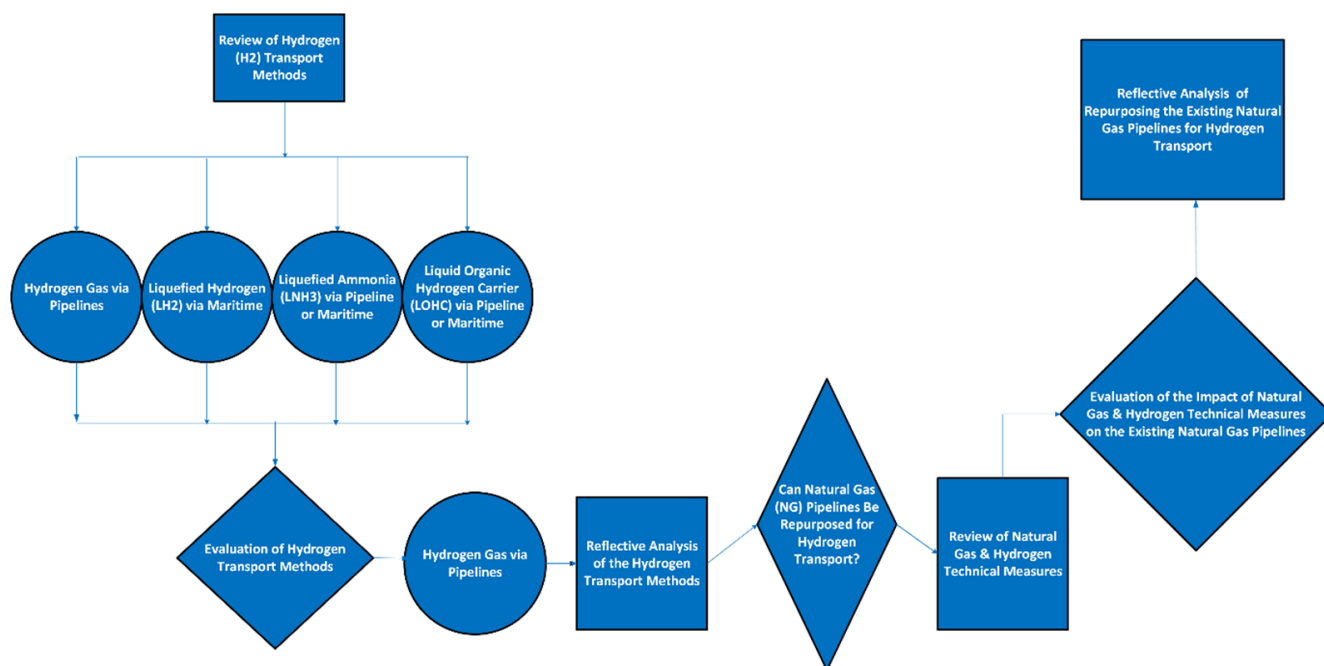
While the literature is replete with studies exploring the feasibility of transporting hydrogen as a blend with natural gas in natural gas pipelines, the transport of pure hydrogen gas has been understudied. Recent research suggests that studies on hydrogen–natural gas blending are more prevalent than those addressing the transportation of pure gaseous hydrogen. The existing natural gas infrastructure can be deployed to transport a blend of hydrogen and natural gas and is widely regarded as a short-term carbon reduction strategy. Hence, there is considerable academic interest in investigating the material suitability, operational safety and performance of the existing NG infrastructure to transport hydrogen–natural gas blend [10,11]. On the other hand, research into pure hydrogen transmission is limited and is generally associated with purpose-built infrastructure which remains limited in deployment and scope of study [12,13].

Consequently, there is a dearth of comprehensive reviews on the impact of technical measures on the existing NG pipelines when repurposed for the transportation of pure hydrogen gas produced from offshore wind farms. This paper addresses these gaps through a comprehensive review of the impact of influential technical measures on the existing natural gas pipeline when repurposed to transport the hydrogen gas produced from offshore wind farms.

## 2. Methodology

As can be seen in Figure 1, a comprehensive review was first completed in this paper to assess all the viable hydrogen transport methods and accordingly identify the optimum transport method by assessing their pros and cons based on key technical characteristics. Twenty-three reputable primary publications were used to carry out this assessment, from which pipelines were identified as the optimum method for the transport of hydrogen gas. Beginning in July 2024, the synthesis was carried out by reviewing the rel-

evant literature, including journal articles and other scholarly publications. Additional sources included published books, white papers, and corporate websites. A significant amount of the literature was obtained from ResearchGate, Academia, ScienceDirect and Google Scholar.



**Figure 1.** Workflow of the study methodology.

The literature review was then further synthesized to assess the impact of the H<sub>2</sub> gas operating characteristics on the existing natural gas pipeline infrastructure, to examine the feasibility of re-using the existing natural gas pipelines for the transport of hydrogen gas. Like the first synthesis conducted to identify the optimum transport option, this stage also involved reviewing an additional twenty-three relevant primary sources, including journal articles and other scholarly publications. Like the first synthesis, a significant portion of the literature was obtained from ResearchGate, Academia, ScienceDirect and Google Scholar.

Overall, for both stages of the synthesis, relevant keywords such as hydrogen (H<sub>2</sub>), H<sub>2</sub> gas transmission, gas pipelines, natural gas transmission, pipeline repurpose, pipeline operating parameters, and transmission costs were used to obtain the most relevant literature. From the materials reviewed, additional relevant sources, particularly older ones, were identified through forward and backward citation searching.

### 3. Review of Existing Hydrogen Transport Methods

#### 3.1. Hydrogen Gas Transport

As the lightest molecule, the density of H<sub>2</sub> is very small. The low density (0.08375 kg/m<sup>3</sup>) and the low volumetric density of gaseous H<sub>2</sub> means that it is challenging to transport gaseous H<sub>2</sub> in large volumes [14–16]. It is imperative to investigate other options to transport H<sub>2</sub>. To boost the gaseous H<sub>2</sub> density and the transportation challenge, compression of gaseous H<sub>2</sub> and transportation via pipelines is an option that has been suggested.

H<sub>2</sub> is not a toxic gas [14] which does not generate much radiation energy [17,18] when it burns. However, with a molecular weight of 2.016 g/mol [19] and density of 0.08375 kg/m<sup>3</sup>, it is small and very light [9,17]. The small size, lightness and flammability make its application challenging [14,19,20]. Because of the lightness, it is more easily leaked through joints, cracks, and seals of containment than other gases [9,17]. Attention should

be paid to the mechanical joints of H<sub>2</sub> transmission lines [18]. Because of this, new valves, fittings, seals, and gaskets [9] should be used for hydrogen transmission lines. Distributed fiber-optic sensing provides continuous, real-time surveillance of hydrogen pipelines by detecting thermal and acoustic anomalies associated with leaks. This technology enables the prompt identification and precise localization of hydrogen release events [21].

H<sub>2</sub> has a wide flammability range which is between 4% and 75% [14]. However, its high flame speed (346 cm/s), low flammability (4%) and low ignition energy (0.02 MJ) [14,22] mean it is explosive [14]. Nonetheless, the impact of any potential fire is mitigated by its low radiation energy and its lightness [14]. On exposure, the lightness enables it to rapidly diffuse through air [14].

H<sub>2</sub> embrittlement occurs when H<sub>2</sub> diffuses on metal surfaces causing the degradation of its mechanical properties [17,19,22,23]. Cracks created by embrittled metals can cause loss of containment and create a flammable atmosphere which can heighten the risk of explosion. Because of embrittlement concerns, H<sub>2</sub> transport pipelines are typically operated between 30 barg and 60 barg [20].

There is significant industrial experience of using H<sub>2</sub>, including its deployment in purpose-built distribution lines with existing site-specific safety practices [14]. In green H<sub>2</sub> production, an electrolyzer is used to generate H<sub>2</sub> from an offshore wind farm [6]. This can then be compressed and transported using pipelines. At 100 barg and 20 °C, the density of compressed gas is  $\approx 7.8 \text{ kg/m}^3$  [20], which is an improvement on the density and the transport capacity of uncompressed gaseous H<sub>2</sub>.

A key feature of transmission by pipelines is the uninterrupted supply of H<sub>2</sub> gas to meet energy demand. It is believed that this is the most economical choice for the long-distance transportation of hydrogen [1]. However, a major handicap is the insufficient pipeline infrastructure, with just 5000 km of H<sub>2</sub> dedicated lines connecting Asia, Europe, and North America, in contrast to the 3 million km for NG [19]. Some of the examples given by Gondal [9] include an 80 km pipeline in Belgium with internal diameter (ID) of 5.9 inches and operating at 100 barg; a 16 km hydrogen pipeline in England, operating at 50 barg; and a 220 km H<sub>2</sub> pipelines in Germany, with ID of 3.9 inches to 11.8 inches, operating at 20 barg.

The low volumetric energy density of hydrogen (10.8 MJ/Sm<sup>3</sup>) presents a transportation challenge [24]. The amount of energy stored per unit volume of a material, fuel, or energy storage system is volumetric energy density [25]. For natural gas, this is 36.4 MJ/Sm<sup>3</sup> [26]. To achieve equivalent energy output, more volumetric flowrates of H<sub>2</sub> must be compressed [27] into bigger diameter pipelines [14]. The high energy density (120.1 MJ/Kg) [14] makes H<sub>2</sub> compression and transport through a pipeline an attractive proposition.

Based on the literature, this is a simple method of transmitting H<sub>2</sub> that guarantees continuous H<sub>2</sub> supply to end users. Its end use includes electricity generation, driving automobiles and industry, or heat, home and business appliances [28]. If this option is utilized, the delivery of non-toxic gas which is dense in energy can be achieved. This is consistent with decarbonization. The high energy density and the continuous supply should compensate for the relatively low volumetric energy and transport densities.

Embrittlement, which increases degradation at higher pressures and concentration, is concerning. Hydrogen embrittlement (HE) arises when gaseous hydrogen penetrates the lattice structure of metal, decreasing its strength and ductility and causing structural cracking at stress levels significantly below the steel's design limit [29,30].

Primarily, hydrogen embrittlement is driven by two mechanisms: the hydrogen-enhanced decohesion (HEDE), in which gaseous hydrogen weakens the atomic bond within the lattice of the steel material, and hydrogen-enhanced localized plasticity (HELP),

where gaseous hydrogen facilitates localized plastic deformation motion, which gives rise to accelerated crack propagation [29,30].

Microstructures significantly influence hydrogen embrittlement. High-strength steels and harder microstructures are more prone to embrittlement due to the likelihood of defects like dislocations and inclusions to trap hydrogen and promote cracking [29,30]. Dislocations are line defects where metal atoms are misaligned, whereas sulfides and oxides which are embedded in metals during production are called inclusions. These give rise to localized structural deformities and elevated stress which can encourage hydrogen accumulation, increasing the vulnerability of the material to cracking [29,30].

Welded joints are particularly prone to hydrogen embrittlement because their thermally altered microstructures can create defects like dislocations and inclusions [31]. In metals, dislocations serve as hydrogen traps allowing hydrogen to accumulate in high-stress zones. This accumulation of hydrogen facilitates crack initiation and accelerates crack propagation [31]. Contaminants, including  $H_2S$  and oxygen, can accelerate cracking. These contaminants can exacerbate cracking through the combined effect of corrosion and embrittlement [32].

There are other safety constraints that can be managed.  $H_2$  is explosive; however, the ease of diffusion of gaseous hydrogen and its low radiation energy inherently mitigates this concern.

### 3.2. Liquefied Hydrogen ( $LH_2$ ) Transport

The low volumetric energy density of gaseous  $H_2$  [27] can create transport issues [16], which can be addressed by liquefying  $H_2$ . Through liquefaction, the density of  $H_2$  can be increased to  $71.1 \text{ kgH}_2/\text{m}^3$  and more volume can be transported [15].

Because of the higher density,  $LH_2$  has good prospects [16,33]. The density ( $71.1 \text{ kgH}_2/\text{m}^3$ ) is superior to gaseous  $H_2$  ( $0.08375 \text{ kg}/\text{m}^3$ ). Like gaseous  $H_2$ , the gravimetric energy density or the energy per unit mass ( $120.1 \text{ MJ}/\text{Kg}$ ) of  $LH_2$  is high [15]. Thus, it is very rich in energy. However, as the boiling point (BP) of  $H_2$  ( $-253^\circ\text{C}$ ) is extremely low, the cooling process to produce  $LH_2$  requires a lot of energy contained [15,33]. An amount of 30 to 44.7% of the energy in  $H_2$  is lost to cooling [15,34,35].

Additional energy losses [15,33] and costs [14,19] are required to conserve the low temperature required during transport in cryogenic vessels [34,35]. Due to this,  $LH_2$  is not deemed efficient for transporting  $H_2$  [19,33]. Also, in transit,  $LH_2$  can be subjected to thermodynamic losses or boil-off gas (BOG) which can impact recovery [15]. BOG is estimated as  $\approx 0.52 \text{ vol\%}$  per day [35]. However, at the destination, evaporation at ambient conditions releases gaseous  $H_2$ , and for this, no energy is lost [15].

### 3.3. Liquefied Ammonia ( $NH_3$ ) Transport

Ammonia has long been utilized as a refrigerant and as feedstocks to produce fertilizers and explosives; customarily, it is shipped in ocean vessels [14]. Furthermore, ammonia contains no carbon molecules, making it a potential candidate for transporting  $H_2$  [36].

Liquid ammonia is transported at a low pressure of 1 barg and below  $-33.34^\circ\text{C}$ , which are considered favorable conditions [33]. Furthermore, ammonia has high auto-ignition levels (15 to 28 vol%), which implies that it will not speedily ignite if an ignition source is nearby. However, ammonia is highly toxic and corrosive—its application has health and safety concerns. Its handling and usage are restricted to competent operating practitioners [14]. Also, ammonia can cause air pollution through acidification if it escapes when partially combusted [14].



Ammonia has a molecular weight of 17.031 g/mol [15]. It requires liquefaction at  $-33\text{ }^{\circ}\text{C}$  [24] to be converted to its denser liquid form for long-distance transport [3] and this liquefaction temperature can be readily achieved [14].

Liquefied ammonia is characterized by high volumetric energy density (14.4 Wh/L) [3]. Because of this, the volume of  $\text{H}_2$  that ammonia can transport is high [14]. Furthermore, the high volumetric  $\text{H}_2$  content ( $121\text{ kg/m}^3$ ) and density ( $686\text{ kg/m}^3$ ) of liquid ammonia [15,33] are good indicators of its capacity to transport  $\text{H}_2$  in large volumes [33].

The BP of ammonia ( $-33.34\text{ }^{\circ}\text{C}$ ) is relatively high, meaning that it is easily liquefied and conserved in liquid form during transport [15]. Because of this, its BOG of 0.024% to 0.1% per day during transport is moderate, meaning that product recovery will be sufficient [15,35]. The BOG (0.024 to 0.1% per day) is less than  $\text{LH}_2$  (0.06 to 0.4%/day), and this equates to higher product recovery [15] compared to the  $\text{LH}_2$  transport method.

Nevertheless, the energy required for dehydrogenation ( $>30.67\text{ MJ/kg}$ ) is massive [15]. Also, comparatively, the gravimetric energy density ( $21.18\text{--}22.5\text{ MJ/Kg}$ ) is low. To put this in context, it is  $120\text{ MJ/kg}$  for  $\text{LH}_2$ .

### 3.4. Liquid Organic Hydrogen Carriers (LOHCs)

LOHCs were first investigated by Japanese researchers conducting studies on Benzene/Cyclohexane systems in the 1980s [37]. Fundamentally, LOHCs are organic molecules that can chemically attach  $\text{H}_2$  to their structure and are of similar characteristics [38].

Methylcyclohexane (MCH), an LOHC derived from toxic Toluene [14], is the best known LOHC [15]. Generally used in the production of organic chemicals, it is a colorless liquid with an odor like Benzene [39]. Dibenzyltoluene and Benzyltoluene are other examples of LOHCs [14,40]. However, this report is based on MCH, the most popular LOHC.

MCH is transported in ambient conditions (1 barg and  $20\text{ to }25\text{ }^{\circ}\text{C}$ ) and is therefore stable and safe for transport [19]. However, MCH is flammable. It has a low range of ignition in air (1.2 to 6.7 vol%) [15]; however, any risk of leakage and fire is significantly reduced due to its stability [38]. Furthermore, MCH is a liquid and less harmful compared to gases which can be inhaled. Thus, its transmission should not cause serious safety issues if adequate measures are installed [14].

At  $101\text{ }^{\circ}\text{C}$ , the BP of MCH is high, therefore there is no liquefaction required for transport and thus no energy costs are incurred [19]. However, its gravimetric ( $7.35\text{ MJ/Kg}$ ) and volumetric ( $5.66\text{ Wh/L}$ ) energy densities are low. In addition, its gravimetric (6.1%) and volumetric  $\text{H}_2$  (47.1%) content are also low. Because of this, if used as an  $\text{H}_2$  carrier, LOHC will deliver small amount of  $\text{H}_2$  [15].

At the destination, a significant amount of energy ( $>43.4\text{ MJ/kg}$ ) is required for dehydrogenation [15]. Energy consumed for this process is  $\approx 30\text{--}40\%$  of the energy stored in the  $\text{H}_2$  [41]. Also, following dehydrogenation, residual LOHC molecules must be recycled for hydrogenation [14]. LOHC molecules are costly and should be re-used; however, recycling unloaded LOHCs is complex and costly [14].

### 3.5. Methanol Transport

Methanol ( $\text{CH}_3\text{OH}$ ) is an  $\text{H}_2$  carrier viewed as having the potential to transport  $\text{H}_2$  [33]. It is rich in  $\text{H}_2$  and can be reformed to  $\text{H}_2$  [34]. However, when methanol decomposes to release gaseous  $\text{H}_2$ ,  $\text{CO}_2$  and  $\text{CO}$  are produced [15,33].  $\text{CO}$ , which is very poisonous, is produced [15,33]. Therefore, the use of methanol for  $\text{H}_2$  transport will lead to environmental pollution [14,33]. Based on this, methanol as a hydrogen transport option is not discussed further in this paper.

#### 4. Evaluation of the Hydrogen Transport Methods

To enable the evaluation of the different H<sub>2</sub> transport methods, Table 1 has been developed to summarize the key attributes of the reviewed hydrogen transport options.

**Table 1.** Key characteristics of the different H<sub>2</sub> transport methods [14,15,19,20,27,33].

|                                                      | H <sub>2</sub> Gas | LH <sub>2</sub> | NH <sub>3</sub> | LOHC (MCH)       |
|------------------------------------------------------|--------------------|-----------------|-----------------|------------------|
| Molecular Weight (Wt.), g/mol                        | 2.016              | 2.016           | 17.031          | 98.186           |
| Density in normal conditions, Kg/m <sup>3</sup>      | 0.08375            | 0.08375         | 0.73            | 866.9            |
| Melting point, C                                     | −259.16            | −259.16         | −77.73          | −126.3           |
| Explosive limit in air, vol%                         | 4 to 75            | 4 to 75         | 15 to 28        | 1.2 to 6.7       |
| Flame speed, cm/s                                    | 346                | 346             | -               | -                |
| Liquefaction energy, MJ/Kg                           | -                  | 15.1 to 57      | >6.73           | -                |
| Transport pressure, Barg                             | 100                | 1.013           | 1.013           | 1.013            |
| Transport temp, C                                    | 20                 | −252.87         | −33.34          | 20 to 25         |
| Density in transport conditions, kg/m <sup>3</sup>   | 7.8                | 71.1            | 686             | 866.9            |
| BOG, % day                                           | -                  | 0.06 to 0.4     | 0.024 to 0.1    | 0.00416 to 0.065 |
| Gravimetric energy density, MJ/Kg                    | 120                | 120             | 21.18 to 22.5   | 7.35             |
| Gravimetric H <sub>2</sub> content %                 | 100                | 100             | 17.8            | 6.1              |
| Volumetric energy density, MJ/Nm <sup>3</sup> [Wh/L] | 13 [8.49]          | 13 [8.49]       | [12.92 to 14.4] | [5.66]           |
| Volumetric H <sub>2</sub> content kg/m <sup>3</sup>  | 100                | 70.8            | 121             | 47.1             |
| Dehydrogenation energy (MJ/Kg)                       | -                  | -               | 30.67           | >43.4            |

The H<sub>2</sub> transport methods reviewed were then analyzed to evaluate the pros and cons of each method, and Table 2 was developed to demonstrate this evaluation. The characteristics in Table 1 were compiled based on their frequent application in the assessments of hydrogen transport systems. Attributes such as volumetric energy density, gravimetric energy density, molecular weight, operating pressure, liquefaction energy, dehydrogenation energy, boiling point and gas boil-off were extracted from the literature. Table 1 consolidates key technical data for comparative analysis, rather than providing a systematic classification of the literature. The primary aim of Table 1 is to highlight important decision-making attributes found in the literature.

Based on Table 1, Table 2 compares transport methods by analyzing the literature for performance, energy losses and operational constraints. Using a uniform, non-weight approach, this qualitative evaluation aims to identify the most practical and cost-effective hydrogen transport option. Following the detailed literature review and the analysis of the identified hydrogen transport methods, as summarized in Table 2, the transport of hydrogen as a gas via pipelines has been determined as the optimum option, offering the highest benefits with minimum drawbacks. The high energy density and the continuous supply characteristic of this option make it the most viable choice. It is also cost efficient as it requires no liquefaction nor dehydrogenation before utilization, thus eliminating the associated energy costs.

**Table 2.** Assessment of the Hydrogen Transport Methods.

| H <sub>2</sub> Transport Methods | Pros                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Cons                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H <sub>2</sub> Gas in Pipelines  | <ul style="list-style-type: none"> <li>• Very clean, non-toxic fuel that when it burns, no GHG is produced.</li> <li>• Low radiation energy and diffuses easily.</li> <li>• Very rich in hydrogen energy. It has very high energy density.</li> <li>• Requires no liquefaction, and therefore no associated energy cost.</li> <li>• No requirement for dehydrogenation and hence, no energy cost.</li> <li>• Boil-off gas (BOG) is non-existent.</li> <li>• Continuous delivery of H<sub>2</sub></li> <li>• Transport density can be increased by compression</li> </ul> | <ul style="list-style-type: none"> <li>• Low auto-ignition temperature and high flame speed and therefore very explosive.</li> <li>• Low volumetric energy and transport density.</li> <li>• Burns in air with invisible flames and thus may not be quickly detected</li> </ul>                                                                                                                                                                                                                                                            |
| LH <sub>2</sub>                  | <ul style="list-style-type: none"> <li>• LH<sub>2</sub> is a very clean, non-toxic fuel.</li> <li>• No energy consumption required to release gaseous H<sub>2</sub> at the destination.</li> <li>• It is very rich in hydrogen energy, like compressed H<sub>2</sub>.</li> </ul>                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>• Low auto-ignition temperature and high flame speed and therefore very explosive.</li> <li>• Extremely energy intensive and wastes too much energy. Cooling consumes 30 to 36% of energy contained in H<sub>2</sub>.</li> <li>• May require significant investment due to the low temperature conservation requirement.</li> <li>• Significant gas boil-off.</li> <li>• Low volumetric energy and storage densities.</li> </ul>                                                                    |
| Liquid Ammonia                   | <ul style="list-style-type: none"> <li>• High auto-ignition temperature and less explosive than H<sub>2</sub>.</li> <li>• Does not produce CO<sub>2</sub>, a GHG when oxidized.</li> <li>• High volumetric energy and transport density. It will deliver high volume of H<sub>2</sub>.</li> <li>• Boiling point is −33.34 °C. Its liquefaction, storage, and conservation in liquid state demands less energy than LH<sub>2</sub>.</li> <li>• Transportation is at ambient pressure.</li> <li>• Cooling requirement of &lt;−33.34 °C can be readily achieved.</li> </ul> | <ul style="list-style-type: none"> <li>• Un-combusted ammonia could cause pollution via acidification.</li> <li>• It is highly toxic and corrosive.</li> <li>• It is not rich in hydrogen energy. The gravimetric energy density is low.</li> <li>• Some gas boils off.</li> <li>• Significant amount of energy is required for dehydrogenation.</li> </ul>                                                                                                                                                                                |
| LOHC (MCH)                       | <ul style="list-style-type: none"> <li>• LOHCs, including MCH, are stored and transported at ambient pressure conditions and are therefore stable and safe to transport.</li> <li>• MCH is a liquid and less harmful than gases which can be inhaled.</li> <li>• No liquefaction required, and therefore no associated energy cost.</li> <li>• It is stable and a liquid. Pipelines and maritime can be used for transport.</li> </ul>                                                                                                                                   | <ul style="list-style-type: none"> <li>• LOHC (MCH) is very flammable and has serious potential to cause fire.</li> <li>• LOHCs, including MCH, have the least gravimetric and volumetric energy density. Also, it has low volumetric H<sub>2</sub> content. It will deliver low quantity of H<sub>2</sub>.</li> <li>• Requires significant amounts of energy to release gaseous H<sub>2</sub>.</li> <li>• Complex due to requirements to recycle unloaded carriers for hydrogenation, and therefore may not be cost-effective.</li> </ul> |

#### 4.1. Reflective Analysis on Hydrogen Transport Methods

Gaseous H<sub>2</sub> is viewed as a viable solution to decarbonize and realize net zero emission [3,14,19]. However, as reviewed in this work (Table 2), H<sub>2</sub> gas has certain characteristics that can compromise the operational safety of the transmission pipelines made of steel materials. Gaseous H<sub>2</sub> is very volatile; it is very light and has low volumetric density, and because of this, it is difficult to transport it in large volumes. Therefore, it is necessary to assess all the H<sub>2</sub> transport options. The assessed options included transporting H<sub>2</sub> as gas via pipelines, as liquefied H<sub>2</sub> via maritime, as liquid ammonia via pipelines or maritime, as LOHCs via pipelines or maritime and as methanol via pipelines or maritime. After a



detailed review and analysis of the pros and cons of each option, as summarized in Table 2, it has been found that the transport of  $H_2$  as gas via pipelines is the optimum option for hydrogen transport, particularly over short to medium distances, and where existing NG pipelines can be utilized to transport  $H_2$ . The high energy density characteristic of this option and the continuous supply attribute compensate for its relatively low volumetric energy density attribute. Although  $H_2$  is explosive, its ease of diffusion and low radiation energy is a mitigation. Due to hydrogen's wide flammability range and low ignition energy, robust safety measures are required following a leak to prevent fires and explosions. These include rapid leak detection, system isolation and automatic pipeline shutdown, as well as adequate ventilation and continuous monitoring to reduce risk and ensure safe operation [14,42].

The  $LH_2$  transport method is unattractive due to its extremely low cryogenic temperature requirements. While  $LH_2$  transport has high gravimetric energy density like the  $H_2$  gas transport in pipelines, and an even higher transport density, the extremely low cryogenic temperature required for its transport and the excessive boil-off means that this option is very unattractive.

Liquid ammonia shipping offers the best option for the transport of  $H_2$  over long distances as it is rare to find pipelines existing across continents and it is un-economical to build new  $H_2$  gas transport pipelines for such distances. Liquid ammonia additionally offers good transport quality (volumetric energy density, volumetric  $H_2$  content and gravimetric energy density) and can be transported at ambient pressure and a temperature of  $-33.34\text{ }^{\circ}\text{C}$  that can be readily achieved. However, the excessive energy required for dehydrogenation is a major constraint for this method of  $H_2$  transport.

The stability of the liquid organic hydrogen carriers (LOHCs) method at ambient conditions means that they can safely transport  $H_2$  via pipelines or maritime. However, this method is unattractive due to the excessive energy required for dehydrogenation and the need to recycle unloaded molecules. Furthermore, the poor transport quality (volumetric energy density, volumetric  $H_2$  content and gravimetric energy density) of this transport method means that if it is adopted, it will deliver a very poor quantity of  $H_2$ .

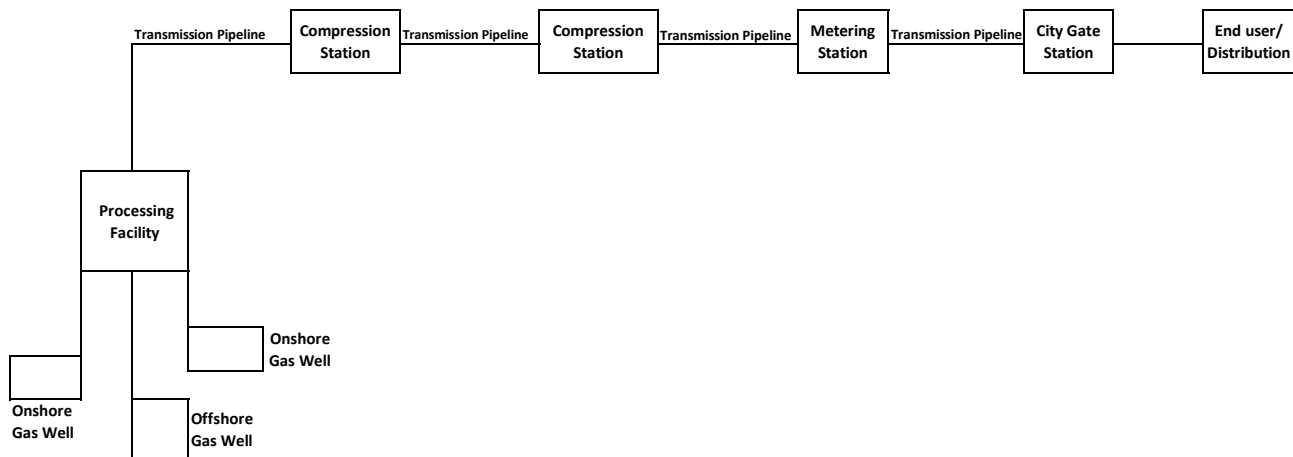
The methanol transport option is unattractive because its decomposition releases poisonous CO and  $CO_2$ . Its technical and safety characteristics were not covered.

#### 4.2. Review of the Existing Natural Gas (NG) Infrastructure

The transport of hydrogen gas in pipelines was identified in the previous section as the optimum method for transporting hydrogen in large quantities, particularly over short to medium distances where the existing NG pipelines can be utilized. In this section, the suitability of the existing oil and gas infrastructure will be assessed with a focus on the existing NG pipelines.

##### The Natural Gas Infrastructure

Figure 2 demonstrates a simplified schematic of an NG transmission system. A typical NG infrastructure consists of pipelines, compression stations [9,27], city gate stations and metering stations that can be installed at intervals on the transmission lines to enable the pipeline operators to monitor and measure the NG in the pipeline and in storage facilities [43].



**Figure 2.** Natural gas transmission system.

The Natural Gas (NG) transmission pipelines typically range between 6 and 48 inches in diameter to transport gas at operating pressures between 10 and 120 barg over long distances and 0.5-inch diameter pipelines are used in the gathering and distribution systems [24,43]. The gas travels through the pipelines when there is a difference in pressure (DP) between two points in a pipeline system. The flow of NG, which must be adequate to meet energy demand, is dictated by the DP in the pipeline.

As NG must maintain pressure while traveling through transmission lines, compressor stations are used to maintain pressures in transmission lines and are usually installed at intervals of 64.4 km to 161 km [43]. Metering stations are installed at regular intervals on the transmission lines to enable the pipeline operators and gas distribution companies to monitor and measure the natural gas flowrates [43]. City gate stations are used to receive the NG from the transmission line before feeding it to the distribution systems. Primarily, the city gate will meter the gas from the transmission pipelines and reduce the pressure to what the distribution system can accept [43].

## 5. Repurposing the Existing NG Pipelines for Hydrogen Transport

In this section, the feasibility of reusing the existing natural gas pipeline infrastructure to transport hydrogen is reviewed and evaluated. The material of the NG pipeline is assessed first; then, the impact of influential technical measures such as gas composition, operating pressure and temperature, volumetric energy density, operating density, and pressure drop on the existing NG pipelines are assessed.

### 5.1. Review of Technical Measures

To establish the possibility of reusing the NG infrastructure for H<sub>2</sub> transport, the material of the NG pipeline is assessed first, followed by an assessment of the effects of technical measures such as the gas composition, operating pressure, operating temperature, volumetric energy density, operating density, and the pressure drop, on the safe and efficient operation of the gas pipeline.

#### 5.1.1. Assessing the Pipeline Materials

Due to the NG high pressures involved [9], most NG pipelines are made from carbon steel [9] or stainless steel [24]. Material grades such as X52, X56, X60, X65, X70 and X80, which are high-strength steels, are typically employed for NG pipelines [44,45]. While high-strength steels such as X52, X56 and X60 have a maximum operating pressure (MOP) of 207 barg, X65, X70 and X80 have a maximum operating pressure of 103 barg [45].

High-strength steels like X100 and X120 material grades are still under research and development, and their application in major projects is still challenging because the very high strength of these material grades makes them very prone to hydrogen-induced stress corrosion-cracking and failure [45]. Common steel grades for natural gas distribution lines [24] are lower strength material grades A, B, X42 and X46 [44,45]. Material grades A, B, and X42 have a maximum operating pressure (MOP) of 207 barg [45].

In hydrogen pipelines, operating pressure levels are generally classified as low (<10 barg), medium (10 to 100 barg), and high (>100 barg) [46,47]. Lower strength API 5L Grade B, X42, X46, and X52 are generally utilized for low- or medium-pressure hydrogen pipelines because they typically show higher ductility and greater resistance to hydrogen-induced cracking [48,49]. Higher-strength pipeline steels such as X60, X65, X70, and X80 are generally utilized in high-pressure natural gas transmission systems because they enable increased throughput and improved transmission efficiency [46]. However, higher-strength steels are more vulnerable to hydrogen embrittlement, hydrogen-induced cracking and crack initiation, especially under fluctuating pressure conditions [50].

Currently, polyethylene (PE) materials are more common in the NG distribution pipelines [9]. PE pipes are now used in lieu of old iron and steel pipes [44]. As per the American Water Works Association, PE material is typically rated for just 17.5 barg [51], with no evidence to suggest that it is prone to hydrogen embrittlement [52]. From a safety and economic perspective, PE is acceptable as a pipeline material to transport hydrogen [20].

The existing H<sub>2</sub> transmission pipelines operate between 30 barg and 60 barg utilizing lower strength steels which are usually API 5L X42 or below [20]. According to Mohitpour et al. (2017) [48], under these operating conditions, little to no failures have been experienced [20]. In addition to this, according to Khan, M.A., Young, C. and Layzell, to contain the threat of H<sub>2</sub> embrittlement, under normal operating conditions, low-strength material grades A, B, X42 and X46 are employed for H<sub>2</sub> pipelines [24]. Furthermore, the American Society of Mechanical Engineers (ASME) [45] declares that API grades below X42 and X52 are less affected by H<sub>2</sub> embrittlement [22] and are approved for hydrogen pipelines.

Steels constructed from lower material grades are better suited to transporting hydrogen since lower strength material grades are less vulnerable to hydrogen-induced cracking. Higher strength material grades have more dislocations and microstructural defects that have the capacity for trapping hydrogen and initiating crack propagating [30,53,54]. By contrast, low-grade pipeline steels have less localized internal stress, are generally more ductile and show reduced susceptibility to hydrogen-induced cracking [53,54]. In general, lower strength steel materials like X42-X52 retain good ductility when exposed to hydrogen, and if welds and microstructures can be managed, like natural gas pipelines, they can be operated at elevated pressures up to 140 barg [55]. Although it is economically more beneficial to use high-strength steel pipelines for hydrogen transport instead of low-strength steel because of its smaller pipe thickness [46], high-strength steels which are often used for NG transmission are more prone to H<sub>2</sub> embrittlement [24]. Based on experiments, the ductility and the strength of higher strength steel grades like X65, X70 and X80+ reduce with increasing hydrogen concentration and under high-stress conditions and are therefore more prone to embrittlement [56,57]. The thermally altered zones of higher strength metals are particularly vulnerable to hydrogen embrittlement [57]. To manage hydrogen embrittlement, these higher strength material grades are made to undergo heat treatment or have their operating pressures reduced [57]. Due to the increased risk of hydrogen-induced cracking, higher strength pipelines operating in hydrogen environments are generally operated at reduced pressures compared with natural gas pipelines [55].

Furthermore, with major modifications, the existing NG pipelines constructed from higher strength steel grades can be utilized to transport pure H<sub>2</sub>. Degradation inhibitors

such as oxygen (O<sub>2</sub>), sulphur (IV) oxide (SO<sub>2</sub>), and carbon monoxide (CO) and pipe-in-pipe technology can be used to modify the existing NG pipelines constructed from higher strength steels [58]. The pipe-in-pipe technique can be used for the modification of the existing NG pipeline by inserting an inner pipeline of material that is not prone to hydrogen embrittlement to provide a physical barrier between the steel pipeline and the gaseous hydrogen being transported [58].

In general, for the re-use of the existing natural gas for the transportation of hydrogen gas, the American Society of Mechanical Engineers B31.12 [45] recommends that the material grade, weld quality, fracture toughness and operating pressure of the repurposed pipelines be assessed before repurposing. As part of these assessments, ASME advises utilizing in-line inspection (ILI) to assess the pipeline condition prior to repurposing.

Therefore, based on the undertaken review, it can be concluded that using NG pipelines that are made of material grades X52 and below to transport H<sub>2</sub> at reduced pressures of 30 barg to 60 barg will reduce the threat of H<sub>2</sub> embrittlement [20,22,24].

### 5.1.2. Gas Composition

To meet energy requirements, the NG flowing in the NG pipelines consists of roughly 87 to 96% methane (CH<sub>4</sub>) [44]. With methane being the main component of the NG [17], the typical composition of the NG flowing in NG pipelines [10] is given in Table 3 [1]. NG whose primary component is methane has a high volumetric energy density (35.8 MJ/Sm<sup>3</sup>) which is thrice the volumetric energy density of hydrogen (10.8 MJ/Sm<sup>3</sup>) [24].

**Table 3.** Natural Gas Typical Composition [1].

| Component      | Mol Fraction (%) |
|----------------|------------------|
| Methane        | 93.76            |
| Ethane         | 3.14             |
| Propane        | 0.62             |
| Butane         | 0.2              |
| Pentane        | 0.07             |
| Nitrogen       | 2.03             |
| Carbon dioxide | 0.18             |

Any change in the natural gas composition could create difficulties in the pipeline operation because this change can affect gas (mixture) properties such as density, dynamic viscosity, Joule–Thomson coefficient, heat capacity, thermal conductivity, volumetric energy density, causing metering inaccuracy [10].

As a transitional measure, the UK is targeting 20 to 30% of hydrogen in a natural gas mixture [1]. As a permanent measure, it is expected that 100% of hydrogen can be transported through the existing high-pressure pipelines [1].

Hydrogen composition is a determining factor on the degree of pipeline degradation [23]. Some metal pipelines can be compromised and degraded during the prolonged exposure to hydrogen existing at high concentrations and pressures [1]. Hydrogen partial pressure within a pipeline directly influences the integrity of the steel materials. The absorption and diffusion of hydrogen into metals increases at higher hydrogen partial pressure, accelerating hydrogen embrittlement [59,60].

In addition, studies have shown that as the hydrogen blending ratio increases, the hydrogen embrittlement (HE) index increases considerably, signifying a significant degradation of the pipeline steel mechanical properties at elevated concentrations of hydrogen [61]. Generally, steels are more prone to embrittlement at H<sub>2</sub> concentrations greater than 30% [1].

For this reason, H<sub>2</sub> can be transported in NG pipelines at reduced concentration if material embrittlement is a concern [17]. H<sub>2</sub> embrittlement, which occurs when H<sub>2</sub> diffuses on metal surfaces causing the degradation of its mechanical properties [17,19,22,23], can compromise pipeline operational safety. The transportation of gaseous hydrogen through the existing NG pipelines can lead to the rapid diffusion of gaseous hydrogen into the steel lattice, increasing fatigue and promoting embrittlement [45]. Cracks created by embrittled metals can cause loss of containment and create a flammable atmosphere which can heighten the risk of explosion.

Hydrogen embrittlement can be prevented by restricting the ingress of hydrogen into the lattice structure of the host pipeline material [1]. Rigorous assessment of the pipelines' welds and joints can give indication of the likelihood of future H<sub>2</sub> embrittlement and pipeline degradation [33]. Weld quality has a major influence on the integrity of hydrogen pipeline. Since welds and associated heat affected zones (HAZs) differ structurally and microstructurally to their parent metal, they are more vulnerable to hydrogen embrittlement [62,63].

Welds of poor quality—typified by defects, microstructural inhomogeneity, and significant internal stresses—can cause elevated local stresses and facilitate hydrogen accumulation. This increases the possibility of crack formation, thus reducing the tensile strength and crack resistance of welded joints compared to non-defective base metals [62,63].

#### 5.1.3. Operating Pressure

There has been a steady rise in the operating pressure of natural gas pipelines over the years. Based on the reviews undertaken, the operating pressure of the NG transmission system ranges from 10 barg to 138 barg [9,24,44,64] at 37.8 °C to 48.9 °C [44], whereas the maximum operating pressure of a pipeline transporting H<sub>2</sub> is 100 barg and at about 20 °C. At this operating condition of 100 barg and 20 °C, the density of the compressed H<sub>2</sub> gas is about 7.8 kg/m<sup>3</sup> [20].

To maintain the continuous supply of energy, the operating pressure of the pipeline needs to be increased above 100 barg to push more volumetric flow of hydrogen gas [10]; however, operating pipelines at increased pressure could adversely impact the integrity of the NG pipeline if it goes beyond its design pressure [10]. However, the impact of embrittlement is exacerbated by pressure [1,17,23,46]; consequently, H<sub>2</sub> transport pipelines are typically operated between 30 barg and 60 barg [20]. At high pressures, weld defects in the pipeline material may increase the risk of pipeline failure. Consequently, design standards recommend appropriate material selection, crack mitigation measures, and strict inspection protocols to ensure safe operation in high-pressure conditions [45,48].

In addition to high pressure, cyclic pressure can also exacerbate hydrogen embrittlement [65]. The fluctuating operating pressure of hydrogen pipelines caused by pipeline demand imposed on a pipeline can accelerate fatigue crack growth and reduce the fatigue life of the pipeline steel [65]. Cyclic pressure accelerates the diffusion of hydrogen within the metal, thereby promoting crack propagating, causing leakage and loss of hydrogen containment [65–67]. These outline the importance of considering cyclic pressures when assessing the repurposing of existing NG pipelines for hydrogen transport [50,68].

#### 5.1.4. Operating Temperature

Based on the undertaken review, the operating temperature of the natural gas pipeline ranges from −6.7 °C to 60 °C [9], whereas the normal operating temperature of the hydrogen pipeline is below +50 °C [23].

Furthermore, based on the reviews undertaken, it has been found that when the pressure of the NG pipeline is reduced, temperature drops by 0.5 °C for every 1 bar



reduction [9]; this phenom is called the Joule–Thomson (J-T) effect [9,27]. This J-T effect can impact the NG pipeline materials and cause safety issues. Throttling NG from 80 barg to 15 barg causes the temperature to drop by 32.5 °C, which is significant [9].

In NG transmission, the J–T effect can lead to serious shifts in temperature that can cause profound thermodynamic change in the NG pipeline infrastructure. The cooling effect can lead to the formation of hydrates that can plug valves or gas pipelines when water drops out of the wet hydrocarbons [1]. To avoid the formation of ice-like hydrates, which have pipeline safety implications, the NG pipeline is usually heat-traced [9] to prevent hydrocarbon liquids from condensing in the pipeline [10].

However, in H<sub>2</sub> transmission, when pressure is reduced, H<sub>2</sub> temperature increases by 0.035 °C for every 1 bar reduction [9,27]. Reducing the pressure of H<sub>2</sub> from 80 barg down to 15 barg causes a 2 °C rise in temperature which does not have any safety implication [9,23,27] for the existing NG pipeline.

#### 5.1.5. Volumetric Energy Density

NG has three times the volumetric energy density (35.8 MJ/Sm<sup>3</sup>) of hydrogen (10.8 MJ/Sm<sup>3</sup>) [24]. For hydrogen to replace the natural gas in the existing NG infrastructure, the volumetric flowrates of H<sub>2</sub> supplied to the end user must be increased [24] to meet the energy content requirements [9].

However, increasing the volumetric flowrate can cause the flow velocity to increase; therefore, the H<sub>2</sub> flow velocity must be monitored to be kept below 20 m/s (max) when flowing in the repurposed NG pipelines to prevent its internals from eroding.

A few mathematical relationships are used to relate the gas properties to its flowrate, the length of pipe, pipe diameter and the inlet and outlet pressures [69]. Some of these equations are the:

- General energy equation
- AGA equation
- Weymouth equation
- Panhandle A and B

Along with the gas laws, equation 1 [69] is the basic energy equation applied to study the performance of a gas pipeline. If the pressure at the inlet and outlet of a pipeline segment are known, the steady-state isothermal flowrate of the gas through the pipeline can be estimated [9].

$$Q = 38.77 \left( \frac{T_b}{P_b} \right) E \frac{\sqrt{1}}{f_f} \left[ \frac{P_1^2 - P_2^2}{S L_m T_{avg} Z_{avg}} \right]^{0.5} D^{2.5} \quad (1)$$

where:

- Q = flowrate of gas, cubic feet per day at base conditions.
- T<sub>b</sub> = base absolute temperature, °R (ANSI 2530 specification: T<sub>b</sub> = 520°R).
- P<sub>b</sub> = base absolute pressure, psia (ANSI 2530 specification: P<sub>b</sub> = 14.73 psia)
- E = pipeline efficiency factor (fraction).
- F<sub>f</sub> = Fanning friction factor
- P<sub>1</sub> = inlet pressure, psia
- P<sub>2</sub> = outlet pressure, psia
- S = specific gravity of flowing gas (air = 1.0)
- L<sub>m</sub> = length of line, miles.
- T<sub>avg</sub> = average temperature, °R, [T<sub>avg</sub> = 1/2 (T<sub>in</sub> + T<sub>out</sub>)]
- Z<sub>avg</sub> = average compressibility factor
- D = internal diameter of pipe, feet

### 5.1.6. Operating Density

The operating density of NG is nine times that of hydrogen [23]. The high contrast between the density of natural gas (74.62 kg/m<sup>3</sup>) and the density of hydrogen (0.008 kg/m<sup>3</sup>) means that, compared to natural gas, hydrogen travels at a higher velocity in the NG pipelines [1]. The density of natural gas and hydrogen of 74.62 kg/m<sup>3</sup> and 0.008 kg/m<sup>3</sup> respectively are at 100 barg and 303 K [1]. This higher velocity of hydrogen in comparison to NG does not cause any safety concerns unless it reaches the erosional limit above which its contact with the pipeline internal walls increases the pipe's vulnerability to internal erosion and failure [1]. Theoretically, there is no limit for gas velocity in pipelines [1]. Nevertheless, a maximum of 20 m/s is recommended to prevent the erosion of pipelines [1].

The commonly used equation for the determination of erosional velocity is the API RP 14E [70]

$$V_e = \frac{C}{\sqrt{P_m}} \quad (2)$$

where:

- $V_e$  = fluid erosional velocity (feet/s)
- $C$  = empirical constant
- $Q_m$  = gas/liquid mixture at flowing pressure and temperature (lbs./ft<sup>3</sup>)

For continuous service Values of  $c = 100$  and for intermittent service,  $c = 125$  [70]. However, at times, this is considered too conservative and unsuitable because it is designed to be used in the design of new offshore piping systems [10]. Equation (3) is used for determining the gas pipeline's erosional velocity limit [1].

$$V_e = N \frac{C}{\sqrt{P}} \quad (3)$$

- $V_e$  = erosional velocity (m/s)
- $N$  = constant (1.22) to convert equation 2 to metric unit from field unit
- $C$  = the empirical constant (varies from 100 to 250)
- $\rho$  = gas density (kg/m<sup>3</sup>)

Khan, Young and Layzell (2021) [24] also gave Equation (4) to calculate the erosional velocity.

$$V_{\max} = 100 \sqrt{\frac{0.05131 Z R T}{G P}} \quad (4)$$

- $V_{\max}$  is erosional velocity in m/s.
- $P$  is gas pressure in kPa.
- $T$  is gas temperature in K.
- $Z$  is compressibility factor at pipeline conditions and is dimensionless.
- $R$  is ideal gas constant in (8.314 kPa·m<sup>3</sup>/kg·mol ·K).
- $G$  is gas gravity

Theoretically, there is no limit for the gas flow velocity in pipelines; however, a 20 m/s threshold is recommended [1] to avoid internal erosion of pipelines that can be intensified by dust, which is common in gas pipelines [1]. For H<sub>2</sub> to replace natural gas in the existing NG pipeline infrastructure, a transport velocity limit of 20 m/s should therefore be considered.

### 5.1.7. Pressure Drop

Pressure drop is one of the most important parameters that should be considered when designing a pipeline infrastructure [23]. A major cause of the pressure drop in pipelines is the frictional losses exerted by the fluid in transport on the walls of the pipes [23].

Gas travels through pipelines when there is a difference in pressure (DP). The difference in pressure is the change in the total pressure between two points in a pipeline system [27]. The flow of NG must be adequate to meet the energy demand, and this flow is dictated by the pressure drop in the pipeline [27].

The frictional pressure drop is related to the Darcy's frictional factor of the fluid stream, the Reynolds number and the relative roughness of the pipe as given by Equation (5) [1].

$$\frac{1}{\sqrt{f}} = -2\log_{10}\left(\frac{\varepsilon}{3.7D} + \frac{2.51}{Re\sqrt{f}}\right) \quad (5)$$

where:

- $f$  = Darcy's friction factor.
- $Re$  = Reynolds number.
- and  $\varepsilon/D$  = relative roughness of the pipe.
- $f$ ,  $Re$  and  $\varepsilon/D$  are all dimensionless [1].

To deliver the required flowrates, compression stations situated roughly every 100 to 500 km along the pipeline are used to boost the pressure loss due to friction [24]. However, the density of hydrogen is much smaller than that of NG, meaning that the pressure drop is less significant for gaseous  $H_2$  [27]. If the flowrate of hydrogen is fixed at three times the flowrate of NG to deliver the equivalent energy output, the resulting pressure drop is expected to be equal for both the NG and hydrogen [27]. The implication of hydrogen gas having the same pressure drop as natural gas is that a fixed flowrate of hydrogen gas can be transported over longer distances without the requirement for additional gas compression [71].

## 5.2. Evaluating the Feasibility of Repurposing of the Existing NG Pipelines for $H_2$ Transport

Based on the impacts of technical measures reviewed in Section 4, Table 4 has been developed to summarize the findings about  $H_2$  transport against NG transport, thus enabling the evaluation of the feasibility of repurposing the existing natural gas pipelines for  $H_2$  transport.

From Table 4, it can be seen that the main component of the NG transported through the NG pipelines for industrial and commercial applications is methane; therefore, if the NG pipeline is to be repurposed for the transportation of pure hydrogen gas, the hydrogen composition will be a determining factor on the degree of embrittlement [23] where metal pipelines can be compromised and degraded during the prolonged exposure to hydrogen existing at high concentrations and pressures [1]. Rigorous assessment of the pipelines, welds and joints will give an indication of the likelihood of future  $H_2$  embrittlement and pipeline degradation [27].

It can also be seen that the transmission of natural gas in the pipeline infrastructure can be operated at up to 138 barg at operating temperatures between  $-6.7^\circ\text{C}$  and  $60^\circ\text{C}$ , whereas the maximum operating pressure of the pipeline if transporting  $H_2$  is 100 barg at about  $20^\circ\text{C}$  [20]; any increase in the pressure increases embrittlement [1,17,23,72]. Because of the embrittlement concerns,  $H_2$  transport is typically operated between 30 barg and 60 barg utilizing low-strength steel (API 5L A, B, X42, and X46) and X52 material grade pipelines [20,22,24,46].

Regarding the JT Effect, when the pressure of NG is reduced, temperature drops by  $0.5^\circ\text{C}$  for every 1 bar reduction [9,27] and this may impact the pipeline materials. J-T effect in NG transmission can plug transmission materials and cause safety issues. However, in  $H_2$  transmission, there are no safety concerns with the J-T effect [9,23,27] because when the pressure is reduced,  $H_2$  temperature increases by  $0.035^\circ\text{C}$  for every 1 bar reduction [9,27].

**Table 4.** Evaluating the feasibility of repurposing the existing NG pipelines for H<sub>2</sub> transport.

|                           | <b>NG Transport in Pipelines</b>                                                                                                                                                      | <b>H<sub>2</sub> Transport in Repurposed Pipelines</b>                                                                                                        | <b>Implication of H<sub>2</sub> on the Existing NG Pipeline</b>                                                                                                       | <b>Possible Mitigations</b>                                                                                                                                                              |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gas Composition           | The main component of the NG transported through the NG pipelines is methane (CH <sub>4</sub> )                                                                                       | If NG pipelines are repurposed to transport pure H <sub>2</sub> gas, 100% H <sub>2</sub> becomes the major component flowing through the existing NG pipeline | Metal pipelines can be compromised and degraded with prolonged exposure to H <sub>2</sub> gas at high concentrations and pressure                                     | Proactive monitoring and rigorous assessment of the pipelines' welds and joints will help identify any possibility of embrittlement and pipeline degradation                             |
| Operating Pressure        | The NG pipeline infrastructure can be operated up to 138 barg at operating temperatures between 6.7 °C and 60 °C.                                                                     | The maximum operating pressure of a pipeline transporting H <sub>2</sub> gas is 100 barg at about 20 °C                                                       | Pressure increase will impact embrittlement which can compromise operational safety.                                                                                  | H <sub>2</sub> transport should typically be operated between 30 and 100 barg at <50 °C utilizing pipelines of low-strength material grades (<API X46) and high-strength material (X52). |
| Operating Temperature     | Due to JT effect, when the pressure of NG is reduced, temperature drops by 0.5 °C for every 1 bar reduction. JT effect in NG can plug transmission materials and cause safety issues. | In H <sub>2</sub> pipelines, when pressure is reduced, H <sub>2</sub> temperature increases by 0.035 °C for every 1 Bar reduction, thus no JT issues.         | For H <sub>2</sub> transport, JT effect does not negatively impact the existing NG pipelines.                                                                         |                                                                                                                                                                                          |
| Volumetric Energy Density | NG volumetric energy density (35.8 MJ/Sm <sup>3</sup> ) is three times the volumetric energy density of H <sub>2</sub> (10.8 MJ/Sm <sup>3</sup> )                                     | H <sub>2</sub> less volumetric energy density (10.8 MJ/Sm <sup>3</sup> ) is expected to make its flowrate in the NG pipeline much greater.                    | Erosional velocity can be exceeded, with consequences of pipeline erosion and leakage.                                                                                | Maintain the H <sub>2</sub> transport velocity below the erosional velocity.                                                                                                             |
| Operating Density         | NG density is much greater than the H <sub>2</sub> gas density                                                                                                                        | H <sub>2</sub> is the lightest molecule with a very small density, 0.08375 kg/m <sup>3</sup>                                                                  | H <sub>2</sub> low density means that it travels faster compared to NG in gas pipelines with potential safety concern if the higher velocity reaches erosional limit. | Maintain the H <sub>2</sub> transport velocity below the erosional velocity.                                                                                                             |

Volumetric-energy-density-wise, NG is thrice (35.8 MJ/Sm<sup>3</sup>) that of hydrogen (10.8 MJ/Sm<sup>3</sup>) [24] and this low hydrogen volumetric energy density presents a transportation challenge [24]. For H<sub>2</sub> to transport the same amount of energy as NG in the pipeline at the same pressure and temperature conditions, its flowrate velocity is expected to be much greater [24]. This high flowrate can present safety issues as the erosional velocity can be exceeded, with a consequence of pipeline leakage [24]. Therefore, for H<sub>2</sub>

to replace NG in the existing NG pipeline infrastructure, its transport velocity should be maintained below the erosional velocity, above which its contact with the pipeline internal walls increases the pipe's vulnerability to internal erosion and failure [1]. Theoretically, there is no limit for gas flow velocity in pipelines; however, a 20 m/s threshold is suggested to avoid internal erosion of pipelines that can be intensified by dust which is common in gas pipelines [1].

## 6. Reflective Analysis on the Repurposing Existing NG Pipelines for H<sub>2</sub> Transport

Based on the last section's evaluation, this paper demonstrates the possibility of repurposing the existing NG pipeline infrastructure for hydrogen transport. Using technical parameters such as the operating pressure, temperature, pressure drop, volumetric energy density and density as criteria, the possibility of repurposing the existing NG pipeline infrastructure for hydrogen transport was conducted.

The operating pressure and temperature conditions for the transportation of gaseous H<sub>2</sub> is well within the range which the NG pipelines are typically operated. Typically, the existing NG pipeline operates between 10 barg and 138 barg and at temperatures between −6.7 °C and 60 °C when transporting NG; therefore, it can be operated between 30 barg and 100 barg and at less than 50 °C to safely transport gaseous hydrogen.

To contain the threat of pipeline embrittlement when transporting H<sub>2</sub>, material grades of X52 and below are more ideal to use. High-strength steels which offer a substantial cost benefit of 10 to 40% [46] are known to be more prone to H<sub>2</sub> embrittlement [24]. Nonetheless, with major modifications, higher strength steel grades can be utilized to transport pure H<sub>2</sub>. Degradation Inhibitors such as oxygen (O<sub>2</sub>), sulphur (IV) oxide (SO<sub>2</sub>), and carbon monoxide (CO) and pipe-in-pipe technology can be used to modify the existing NG pipelines constructed from higher strength steels [58]. Degradation inhibitors can be injected into the existing pipeline to inhibit the process of H<sub>2</sub> embrittlement. Utilizing the pipe-in-pipe technique, an inner pipeline material that is not prone to embrittlement can be inserted in the existing pipeline to screen the pipeline from hydrogen gas [58]. Another pipeline material that can be used to transport hydrogen is polyethylene (PE) because, based on the reviews undertaken, there is no evidence that polyethylene (PE) materials are susceptible to embrittlement. However, polyethylene (PE) materials are typically rated 17.5 barg and should not be used to transport gaseous hydrogen at elevated pressures.

Pressure drop (DP) should not pose any serious issue for the transport of gaseous H<sub>2</sub> in the existing NG pipelines due to the very low density of H<sub>2</sub> compared to NG. If the flowrate of hydrogen is fixed at three times the flowrate of NG to deliver the same energy, the pressure drop for H<sub>2</sub> gas and NG is expected to be equal [27] and hydrogen gas can be transported over longer distances than expected without the requirement for additional gas compression [71].

While the J-T effect in natural gas transport can plug the pipeline materials and cause safety issues, H<sub>2</sub> transport does not impact the existing NG infrastructure negatively.

The low volumetric energy density of hydrogen presents a transportation challenge because the flow rate of H<sub>2</sub> must be increased to transport the same amount of energy. Increasing the flowrate can cause flow velocity to increase. Therefore, for H<sub>2</sub> to be transported in a repurposed NG pipeline, the flow velocity must be monitored and kept below the erosional velocity to prevent the pipeline internals from eroding.

In conclusion, based on the above assessment criteria, it may be possible to repurpose the existing NG pipeline infrastructure for hydrogen transport.



## 7. Conclusions

Based on the undertaken review of the different options for H<sub>2</sub> transport and based on the assessment of the feasibility of repurposing the existing NG pipelines for H<sub>2</sub> transport using technical criteria, several outcomes that offer encouragement for the repurposing of the existing natural gas pipeline infrastructure emerged.

A major outcome is the obvious advantage of the transport of H<sub>2</sub> as gas in pipelines, particularly for short to medium distances. This option is characterized by the continuous delivery of non-toxic H<sub>2</sub> gas which is highly dense in energy. The high energy density and the continuous supply characteristic of this option make it the most viable choice. However, for hydrogen to replace natural gas in the existing NG infrastructure, the flowrate of hydrogen gas is expected to be fixed. Hydrogen gas must flow three times the flowrate of natural gas to deliver the same energy. Fortunately, the very low density of H<sub>2</sub> compared to NG means that pressure drop (DP) should not pose any serious issue for the transport of gaseous H<sub>2</sub> transport using the existing NG pipelines. Subsequently, this fixed flowrate is expected to allow hydrogen gas to be transported over longer distances without the requirement for additional gas compression.

Maintaining the transport velocity of the hydrogen gas below the erosional velocity is essential to safeguarding the operating and safety integrity of the existing natural gas pipelines. The H<sub>2</sub> transport velocity in the repurposed NG pipeline must be kept below the erosional velocity which has been recommended to be below 20 m/s. Keeping the transport velocity below 20 m/s is critical to preventing the erosion of pipeline internals and avoiding leakage. The distributed fiber-optic sensing technology provides continuous real-time surveillance of hydrogen pipelines, enabling prompt detection and precise localization of hydrogen release events. This can help mitigate the risk of leaks.

Furthermore, unlike natural gas, the J-T Effect does not create operational safety issues if H<sub>2</sub> is transported in repurposed NG pipelines. This attribute of hydrogen gas generates opportunities for the safe transportation of hydrogen gas, an encouraging development for the repurposing of natural gas pipelines for hydrogen gas transportation.

From a pipeline materials perspective, it is possible to repurpose the existing natural gas pipeline for hydrogen gas transportation. The maximum operating pressure of a pipeline transporting H<sub>2</sub> is 100 barg at about 20 °C. As pressure increases the impact of embrittlement, material grades X52 and below are more ideal to use to transport H<sub>2</sub> in NG pipelines at reduced pressures of 30 barg to 60 barg to contain the threat of H<sub>2</sub> embrittlement. Low-strength steel of material grades API 5L A, B, X42, and X46 and X52 are considered not susceptible to hydrogen embrittlement at normal operating conditions. Therefore, with no modification, it is possible to reuse the existing NG pipelines that are constructed from low-strength steel below X46 and X52 to transport pure hydrogen gas.

With major modifications, higher strength material grade (X56, X60, X65, X70, and X80) pipelines can be repurposed to transport pure H<sub>2</sub>. Degradation inhibitors (O<sub>2</sub>, SO<sub>2</sub>, and CO) and pipe-in-pipe techniques can be applied to modify the existing NG pipelines constructed from higher strength steels. Injecting the degradation inhibition gases into the steel material pipeline can prevent gaseous hydrogen from being absorbed into the pipeline material [58]. In the pipe-in-pipe (PIP) technique, an internal pipe is installed into the existing steel pipeline to provide a barrier between the steel pipeline and hydrogen gas. This internal pipe is constructed from a material that can withstand the challenges that the hydrogen gas poses. Polyethylene is the most utilized inner pipe in oil and gas facilities [58]. A key outcome of this study is that the compatibility of existing natural gas pipelines for hydrogen transport depends on factors such as H<sub>2</sub> concentration, pressure cycling, and weld integrity. These factors must be considered in any pipeline repurposing assessment.

Table 5 summarizes the pipeline material grades that can be used for H<sub>2</sub> transport at equivalent operating pressure.

**Table 5.** Material Grades for H<sub>2</sub> Transport.

| Material Grade               | Hydrogen Concentration | Operating Pressure Levels | Associated Risks | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|------------------------|---------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API 5L A, B, API 5L X42, X46 | 100%                   | Low (<10 barg)            | Low              | Higher ductility and greater resistance to hydrogen-induced cracking [48,49].                                                                                                                                                                                                                                                                                                                                                            |
| API 5L A, B, API 5L X42      | 100%                   | Medium (30 to 60 barg)    | Low              | Higher ductility and greater resistance to hydrogen-induced cracking [48,49]. Failures are rare in these operating pressure conditions [48].                                                                                                                                                                                                                                                                                             |
| API 5L A, B, API 5L X42, X46 | 100%                   | Medium (10 to 100 barg)   | Low to Medium    | Higher ductility and greater resistance to hydrogen-induced cracking [48,49]. Employed for H <sub>2</sub> pipelines under normal operating conditions [24]. Normal operating pressure is between 30 barg and 60 barg [20].                                                                                                                                                                                                               |
| API 5L X42-X52               | 100%                   | Medium (10 to 100 barg)   | Low to Medium    | Good ductility when exposed to hydrogen [55]. Reference [45] declares that API grades below X42 and X52 are less affected by H <sub>2</sub> embrittlement and are approved for hydrogen pipelines. Normal operating pressure is between 30 barg and 60 barg [20].                                                                                                                                                                        |
| Polyethylene (PE)            | 100%                   | <17.5 barg                | Low              | Polyethylene (PE) materials are not susceptible to embrittlement [52]. Typically, they are rated 17.5 [51] barg and should not be used to transport gaseous hydrogen at elevated pressures.                                                                                                                                                                                                                                              |
| API 5L X60, X65, X70, X80    | 100%                   | Medium (10 to 100 barg)   | High             | More dislocations and microstructural defects [30,53,54]. Vulnerable to hydrogen embrittlement, hydrogen-induced cracking and crack initiation [50]. Reduced ductility and fatigue strength with increasing hydrogen concentration and high stress [56,57]. Undergo heat treatment or reduce operating pressure to manage the risk of hydrogen embrittlement [57]. It can be re-used with major modifications such as pipe-in-pipe [58]. |

Based on the review conducted in this Paper, practical engineering guidance for hydrogen pipeline repurposing is presented in Table 6.

**Table 6.** Practical engineering guidance for hydrogen pipeline repurposing.

| Technical Guidelines          | Implementation Measures                                            |
|-------------------------------|--------------------------------------------------------------------|
| Material compatibility        | Use of appropriate steel material grade (<X52)                     |
| Operating pressure            | Review operating pressure and derating if necessary                |
| Pipeline modification         | Apply pipe-in-pipe modification technique                          |
| Pipeline integrity assessment | Conduct in-line inspection (ILI) prior to repurposing              |
| Monitoring                    | Apply distributed fiber-optic sensing for real-time leak detection |

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## Abbreviations

|                    |                                               |
|--------------------|-----------------------------------------------|
| H <sub>2</sub>     | Hydrogen                                      |
| NG                 | Natural Gas                                   |
| LH <sub>2</sub>    | Liquified Hydrogen                            |
| BP                 | Boiling Point                                 |
| BOG                | Boil-Off Gas                                  |
| MW                 | Molecular Weight                              |
| LOHC               | Liquid Organic Hydrogen Carrier               |
| MCH                | Methylcyclohexane                             |
| NH <sub>3</sub>    | Ammonia                                       |
| CH <sub>3</sub> OH | Methanol                                      |
| CO <sub>2</sub>    | Carbon Dioxide                                |
| CO                 | Carbon Monoxide                               |
| KM                 | Kilometer                                     |
| DP                 | Pressure Drop                                 |
| J/Scm              | Joules per Standard cubic meter               |
| Btu/Scf            | British thermal unit per Standard cubic meter |
| CH <sub>4</sub>    | Methane                                       |
| JT                 | Joule Thomson                                 |
| PE                 | Polyethylene                                  |
| ASME               | American Society of Mechanical Engineers      |
| O <sub>2</sub>     | Oxygen                                        |
| SO <sub>2</sub>    | Sulphur (IV) Oxide                            |
| °C                 | Degree Celsius                                |
| PIP                | Pipe-in-Pipe                                  |
| MOP                | Maximum Operating Pressure                    |
| GHG                | Green House Gas                               |
| K                  | Kelvin                                        |
| ILI                | In-line Inspection                            |
| MJ/Sm <sup>3</sup> | Mega-Joule per Standard cubic meter           |
| Kg/m <sup>3</sup>  | Kilogram per cubic meter                      |

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